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G2 Amiga release 1.0 · 2026-11-03

Setting Up the Amiga Core on a Pano Logic G2 Revision C

From a factory Pano Logic G2 revision C to a running Amiga: the release files, building and using tnpflash, the stick, the first start, the AmigaOS programs, the settings, updates and the way back to the stock firmware.

HIGHLIGHTS

- One network write with tnpflash; the stock firmware stays in the golden image.
- Firmware from the stick: updates are a file copy.
- Setup on the screen: F12 or a pad's PS button.
- AmigaOS programs from one floppy image with an Install script.

KEY PARAMETERS

PARAMETER	VALUE
Unit	Pano Logic G2 rev C
Flash write	main slot only
Write time	about 700 s
Stick	FAT32 / exFAT
Kickstart	supplied by the user

1 WHAT IS NEEDED

ITEM	REQUIREMENT
Unit	Pano Logic Zero Client G2, revision C (XC6SLX100, M25P64 8 MB flash), with its stock firmware and its power supply. Revision A and B units (XC6SLX150, 16 MB flash) are refused by tnpflash
Network	Ethernet with a DHCP server; the computer on the same network segment as the unit (discovery does not cross a router)
Computer	macOS 11 or later, Linux (x86_64 or arm64) or Windows 7 or later
USB stick	Any size, formatted FAT32 or exFAT
Kickstart ROM	An Amiga Kickstart image of 256 or 512 KB, owned by the user: Kickstart 3.1 (40.068) for the A1200 profile, 1.3 or 2.04 for the A500 profiles. Not part of the release
Disks	AmigaOS or game disk images (*.adf), or a hard-disk image (*.hdf); a new blank hard disk can be made in setup
Monitor	VGA (DVI-I port with an adapter), DVI or HDMI (micro-HDMI port), 640 × 480 at 60 Hz
Input	A USB keyboard and mouse (HID boot protocol, low or full speed); optional: one or two Sony DualShock 4 or DualShock 3 pads

2 RELEASE FILES

FILE	CONTENT
`pano-g2-amiga-1.0-slot.bin`	The flash image for the main slot at 0x380000: the Amiga fabric and its loader. Written by tnpflash
`pano-g2-amiga-1.0-stick.zip`	The stick's \amiga directory: AMIGA.FW, a sample AMIGA.INI, G2TOOLS.adf
`G2TOOLS.adf`	A DD floppy image with the AmigaOS programs and their Install script

FILE	CONTENT
tnpflash-macos	tnpflash for macOS (universal: Apple silicon and Intel)
tnpflash-linux-x86_64	tnpflash for Linux x86_64, statically linked
tnpflash-linux-arm64	tnpflash for Linux arm64 (aarch64), statically linked
tnpflash-windows.exe	tnpflash for Windows (32-bit, runs on 64-bit Windows)
SHA256SUMS	The sha256 of every file

1. Download the files from `https://github.com/mikewolak/pano-g2-amiga/releases/tag/v1.0`.
2. Check them: on macOS and Linux `shasum -a 256 -c SHA256SUMS`; on Windows `certutil -hashfile <file> SHA256` and compare with SHA256SUMS. Every line must read OK.

3 BUILDING TNPFLASH

The release carries tnpflash binaries; building from source gives the same program. The source is the directory `tnpflash/` of the repository and needs only a C99 compiler and make.

3.1 macOS

1. Install the command-line tools: `xcode-select --install`.
2. In the repository: `cd tnpflash && make`. The program is `build/tnpflash`.
3. Run the tests: `make test`. The last line must read `8 passed, 0 failed`.

3.2 Linux

1. Install a compiler: Debian and Ubuntu `sudo apt install build-essential`; Fedora `sudo dnf install gcc make`.
2. `cd tnpflash && make && make test`: `8 passed, 0 failed`.

3.3 Windows

1. Install MSYS2 (msys2.org); in the “MSYS2 UCRT64” shell: `pacman -S mingw-w64-ucrt-x86_64-gcc make`.
2. `cd tnpflash && make win64 MINGW64=gcc`. The program is `build/tnpflash.exe`.
3. Or cross-build on Linux with MinGW-w64: `make win64` (x86_64-w64-mingw32-gcc) or `make win32`.

> The tests run tnpflash against `build/tnpsim`, a simulated G2 revision C on 127.0.0.1: discovery, identification, an 8 MB backup read twice, both refusals of g2write, a write, a verify, and a check that the header, golden image and configuration table are unchanged. Measured: macOS 26 and Ubuntu 24.04 arm64, 8 of 8; the Windows build under wine: info, backup, g2write and verify correct.

4 PREPARING THE STICK

1. Format the stick FAT32 or exFAT.
2. Unpack `pano-g2-amiga-1.0-stick.zip` at the top of the stick: it makes the directory `\amiga`.
3. Copy the Kickstart image into `\amiga` as `KICK_A1200.ROM` (3.1 for the A1200 profile) and, for the A500 profiles, as `KICK.ROM`.
4. Copy the disk images (*.adf, *.hdf) into `\amiga`.
5. Edit `\amiga\AMIGA.INI` if wanted (section 8); every item can also be set later in setup.

```
\amiga\AMIGA.FW          the firmware (release 1.0)
\amiga\AMIGA.INI         settings
\amiga\KICK_A1200.ROM    Kickstart 3.1 (supplied by the user)
\amiga\G2TOOLS.adf       the AmigaOS programs
\amiga\Workbench.adf     disk images (supplied by the user)
\amiga\share\            the USBStick: volume (made at the first start)
```

5 FLASHING THE UNIT

> tnpflash talks to the unit's stock firmware over the network. It writes only the main slot (0x380000–0x6BFFFF); the multiboot header, the golden image (the stock firmware, 0x040000) and the configuration table with the unit's MAC address (0x6C0000) are never written. A unit whose main slot fails to configure starts its golden image, so the stock firmware stays reachable.

1. Connect the unit to the network and power it. The stock firmware takes an address by DHCP.
2. Find it: `tnpflash discover`. Each unit prints its IP address, netmask, gateway, MAC address and revision. With several network interfaces, name the computer's address: `tnpflash discover -l 192.168.1.20`.
3. Identify it: `tnpflash info <ip>`. The board line must read `G2 rev C (LX100, M25P64) -- g2write accepts this unit`.
4. Back up the whole flash: `tnpflash backup <ip> g2-<mac>.bin`. The 8 MB are read twice and compared; the file is written only when both passes agree, with its sha256. Keep this file: it is the only copy of the unit's own flash.
5. Write the Amiga: `tnpflash g2write <ip> g2-<mac>.bin pano-g2-amiga-1.0-slot.bin`. The tool checks the backup against the unit (header and configuration table), erases and programs only the main slot and reads every word back. It ends with `DONE in … s: written and verified` (about 700 s on a wired link).
6. Power-cycle the unit.

MESSAGE	MEANING AND ACTION
nothing answered on UDP 8321	No unit on this segment: check the cable and DHCP; try `-b <subnet broadcast>` or `-l <local ip>`
not a G2 revision C … refusing	A G1 or a G2 revision A/B: this release does not apply
header sector differs / configuration table differs	The backup is from another unit or the flash changed: make a new backup of this unit
WRITE FAILED … DO NOT POWER-CYCLE	The stock image is still running: repeat the g2write command
a stale session it would not reset	`tnpflash reset <ip>`, then repeat

6 FIRST START

1. Insert the stick in any USB socket of the unit, connect the monitor, keyboard and mouse, and power the unit.
2. The loader prints its lines on the screen, mounts the stick and starts `\amiga\AMIGA.FW`; the firmware reads AMIGA.INI, loads the Kickstart and the disks, waits up to 20 s for network time and starts the Amiga.

3. Press F12 (or PS on a pad) to open setup. On page Machine choose Profile A1200 AGA; on page Drives choose the Workbench disk for DF0: or a hard-disk image for HD0:; press F10 and confirm Save and reset.

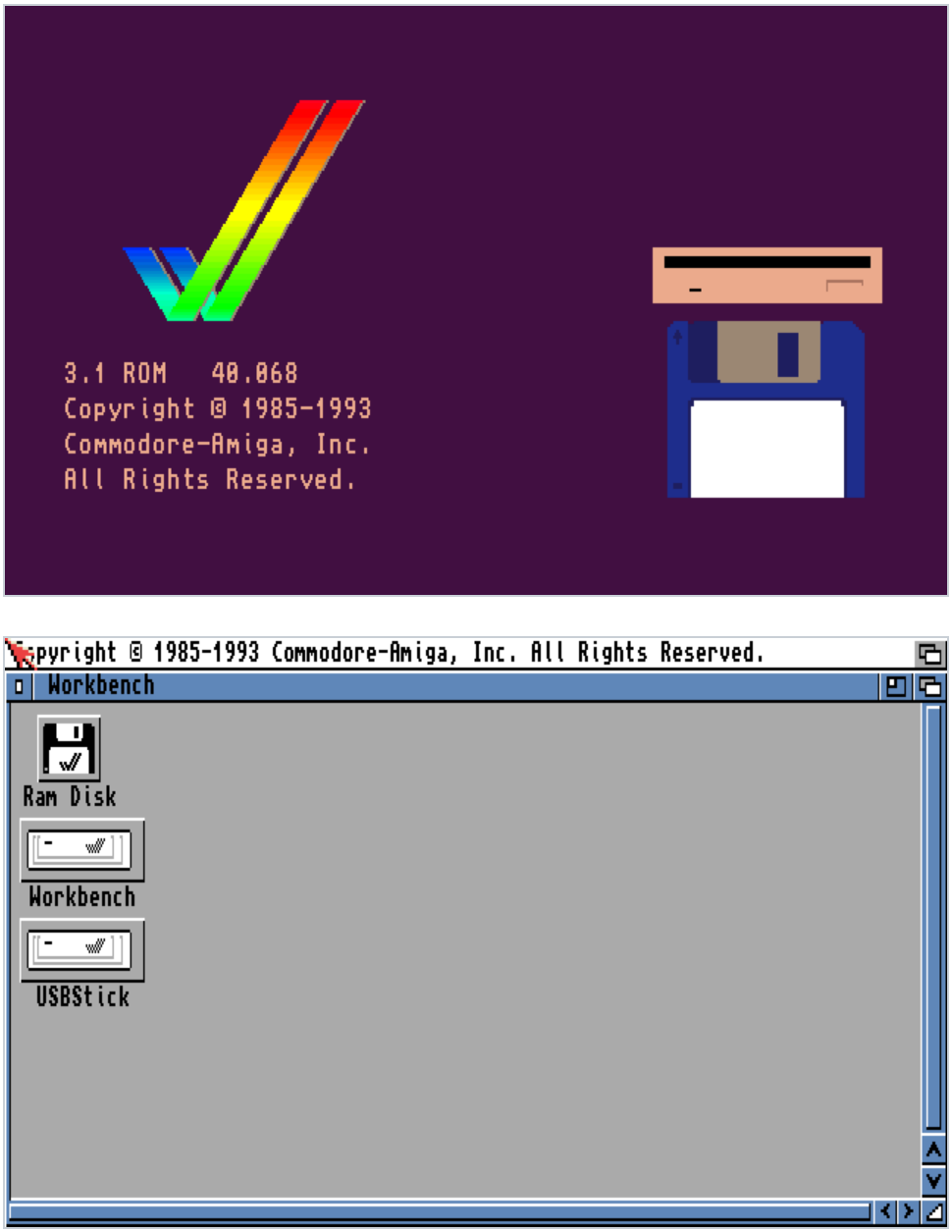


Figure 1. Kickstart 3.1 waiting for a disk, and Workbench 3.1 from a hard-disk image with the USBStick volume, captured from the unit's video.

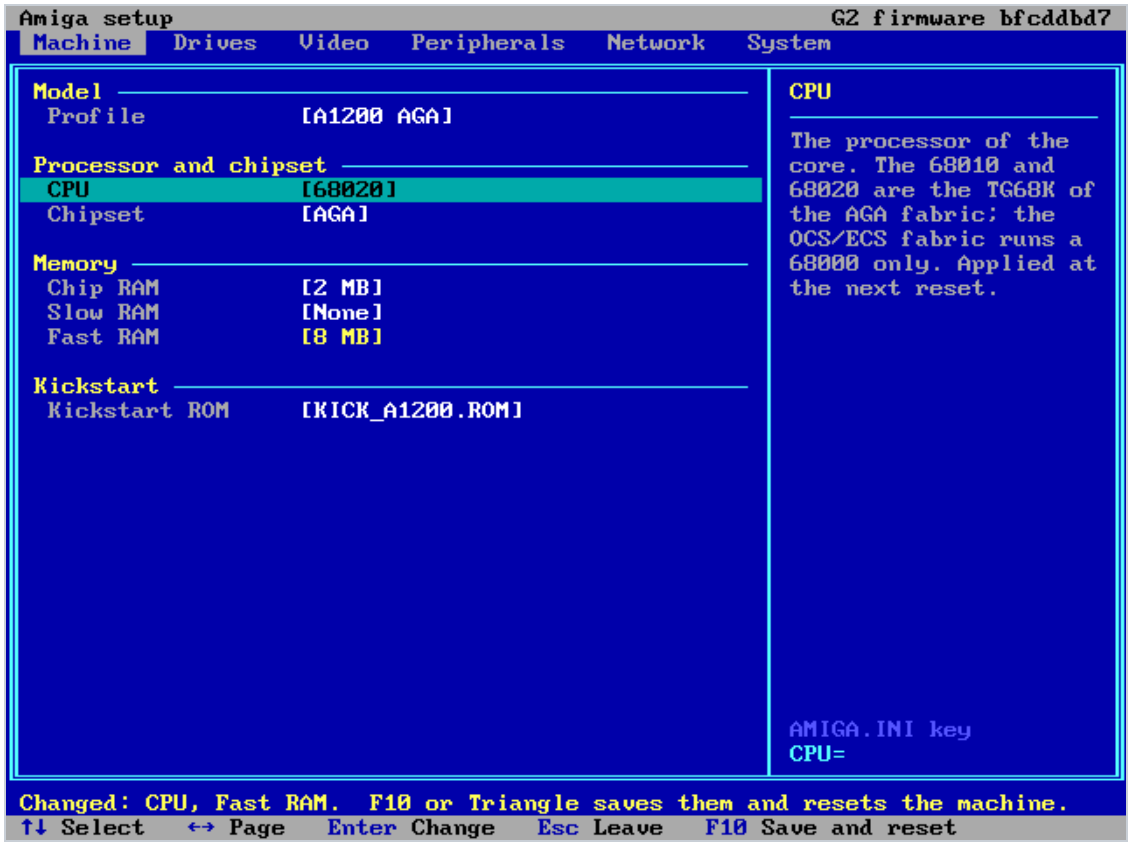


Figure 2. Setup, page Machine. Yellow values are changed and not yet saved; F10 saves them and starts the machine cold.

7 INSTALLING THE AMIGAOS PROGRAMS

1. The sample AMIGA.INI puts `G2TOOLS.adf` in DF1: (DF1=G2TOOLS.adf); with another AMIGA.INI choose it for DF1: in setup, page Drives.
2. In a Workbench Shell: `Execute G2Tools:Install`. The script copies the programs to the system disk (C:, L:, DEVS:) and adds `Run >NIL: C:PANOFS` to S:User-Startup.
3. Reset the Amiga (setup page System, Reset). The volume USBStick appears on Workbench: it is the stick's folder \amiga\share.
4. For AHI sound install AHI 6 (Aminet driver/audio) and choose the G2 Toccata modes in AHI Preferences. For bsdsocket.library programs (FTP, SSH) run `Run >NIL: C:PanoTCP`.

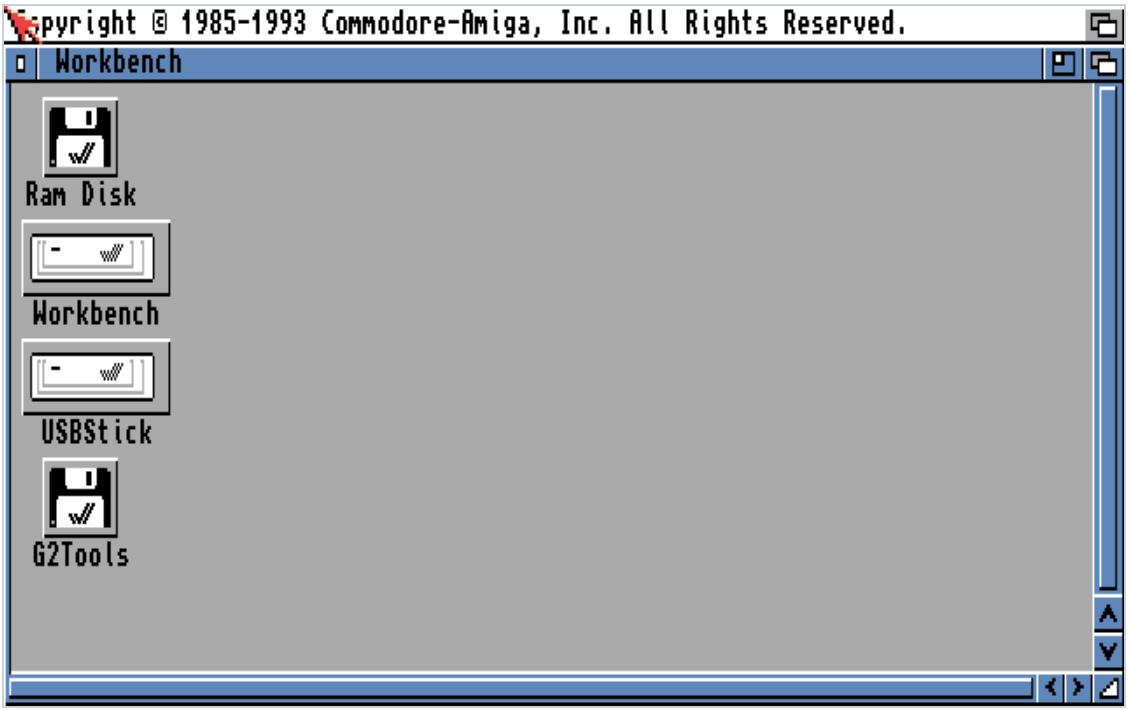


Figure 3. Workbench with G2Tools in DF1: and the USBStick volume. On the test unit the Install script copied the 14 files in 14 s (file dates 13:05:50 to 13:06:04).

8 AMIGA.INI

KEY	VALUES	DEFAULT
MACHINE=	A500, A1200: the defaults of a profile, each overridden by its own key	A500
KICK=	Kickstart image in \amiga	KICK.ROM
CPU= CHIPSET=	68000, 68010, 68020; OCS-A500 (0), OCS-A1000 (1), ECS (2), AGA (3)	68000, ECS
CHIP= SLOW= FAST=	0–3: 512 KB–2 MB; none–1.5 MB; none, 2, 4, 8 MB	3, 1, 0
DF0=…DF3= HD0= HD1=	Image files in \amiga; empty: no disk	DF0: the first *.adf when there is no HD0
WRITE=	0 protects every floppy, 1 writes	1
REGION=	NTSC, PAL	NTSC
VIDEO= SCREEN= HOFS=	SCALE or NATIVE; 0 micro-HDMI, 1 DVI-I, 2 both; 0–400	SCALE, 2, 77
FTDI=	SERIAL, MPSSE	SERIAL
SHARE=	The stick folder of USBStick:	/amiga/share
TZ= NTPWAIT=	Hours east of UTC (-5, +5:30); 0–120 s	0, 20

```
MACHINE=A1200
KICK=KICK_A1200.ROM
DF1=G2TOOLS.adf
REGION=NTSC
VIDEO=SCALE
SCREEN=2
WRITE=1
TZ=0
```

9 UPDATING AND RETURNING TO THE STOCK FIRMWARE

9.1 A new firmware

1. Copy the new `AMIGA.FW` to \amiga on the stick and power-cycle the unit. The flash is written only for a new fabric.

9.2 A new fabric

1. In setup, page System, choose Golden image: the unit restarts into its stock firmware.
2. Run `tnpflash g2write` with the new slot image, as in section 5.

9.3 Back to the stock firmware

1. In setup choose Golden image; the stock firmware runs.
2. Cut the stock main slot out of the backup: `dd if=g2-<mac>.bin of=stock-slot.bin bs=65536 skip=56 count=52`.
3. Write it back: `tnpflash g2write <ip> g2-<mac>.bin stock-slot.bin`, then power-cycle. Untested in this release.

10 TROUBLESHOOTING

SYMPTOM	CAUSE AND ACTION
The loader stops at “no USB stick” or “AMIGA.FW”	The stick is not FAT32/exFAT or \amiga\AMIGA.FW is missing or damaged: copy it again
“Kickstart” error in the boot lines	KICK= names a file that is not in \amiga or is not 256/512 KB
Keyboard or mouse not working	Only HID boot-protocol keyboards and mice are used; \amiga\BOOT.LOG names every device found or rejected
The picture is shifted	Setup, page Video, Picture offset (77 is the standard window)
A PAL game shows its bottom lines cut	Set Region PAL in setup
The unit starts the Pano Logic screen instead of the Amiga	The main slot did not configure: the golden image runs. Repeat section 5

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Repository mikewolak/pano-g2-amiga, release 1.0.